

IT Social and professional problems

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44% Assistance

51% Exam

$$\text{Entropy (Information)} \rightarrow dS = \frac{\delta W}{T} = \left[\frac{J}{K} \right]_{\text{Kelvin}}$$

of different values

$$\text{Quantity of info} = H(x) = - \sum_{i=1}^N P(x_i) \cdot \log_a P(x_i)$$

$a=2 \rightarrow \text{bits}$
 $a = \text{\# different values possible}$

Product - everything that can be offered to the market.

IT product - a product that incorporates IT solutions (product with license)

Ownership = A property right that allows to use and dispose of a thing to the exclusion of people other than the owner.

Co-ownership:

- Joint
- In fractional parts

Presumption of equal distribution

Laws importance:

1. National Laws (Acts in Poland - Amendment in USA) (Most important law)
2. EU Law
3. International Laws

Subject matter of copyright = Any manifestation of creative activity of individual character (determined in any form) regardless of value, purpose ...

A fiction Formula is not creative
A drawing - Yes

Permission must be written in paper not oral.

Subjects that are not copyright:

- Normative acts
- Official documents
- Published patents or protective descriptions
- Simple press releases

Author is not required to sign/register his works somewhere.

The consent of all co-authors is needed to exercise copyright in the entire work.

Permitted use: The scope of personal use includes the use of single copies of works by a circle of people (family and close friends)

Use of work under Fair use: The permitted use may not infringe upon the normal use of the work or harm the legitimate interests of the author.

Scientific and educational institutions may make copies of fragments of the distributed work

Agreement to transfer economic right of the author must be written else is invalid/null/void

Practice exam:

1. Which of the following is non-copyrighted?

- A) Source code with original structure
- B) An algorithm described in pseudocode as a method
- C) A creative GUI layout that is unique
- D) A novel user manual written in original prose

2. Who by default owns the economic rights to a program written by an employee during their work duties?

- A) The employee
- B) The employer
- C) The state
- D) The software collecting society

3. A license agreement that does not state duration is treated as:

- A) Invalid
- B) For 5 years only
- C) Indefinite but terminable by author with notice
- D) Automatically transferable to third parties

4. Which action can be criminally punished under the lecture's list of offences?

- A) Citing a small excerpt from a report with attribution
- B) Unauthorized circumvention of technical protection to copy works for profit
- C) Using ideas learned in class to build new software
- D) Posting a link to a public domain work

5. Duration of economic right after the author's death is typically:

- A) 20 years
- B) 50 years
- C) 70 years
- D) Indefinite

6. Which is NOT an object of industrial property?

- A) Patent for an apparatus
- B) Trademark
- C) A mathematical theorem
- D) Topography of integrated circuits

7. Co-authors' shares are presumed to be:

- A) Proportional to income produced
- B) Equal unless agreed otherwise
- C) Assigned exclusively to the lead author
- D) Owned by the publisher only

8. Which is allowed under permitted use described?

- A) Republishing a textbook chapter online for a public audience
- B) Making a single copy of a distributed work for personal use within a family circle
- C) Selling scanned copies of a journal article for profit
- D) Using a database commercially without permission

9. Which of the following describes the legal status of computer programs?

- A) Unprotected in all forms
- B) Protected as literary works
- C) Only hardware is protected, not software
- D) Protected only if registered with government

10. A professional code of ethics for software engineers includes:

- A) Acting in the public interest and avoiding conflicts of interest
- B) Always prioritizing employer profit over public safety
- C) Keeping all professional knowledge secret forever
- D) Encouraging use of unlicensed software to reduce costs

Solution:

- 1 - B
- 2 - B
- 3 - C
- 4 - B
- 5 - C
- 6 - C
- 7 - B
- 8 - B
- 9 - B
- 10 - A

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Core legal concepts:

- Intellectual Property (IP) is divided in:
 - Copyright: Protection of original expression: work
 - industrial property: Patents, utility models, industrial designs, trademarks, topographies of integrated circuits).Legal sources include national law, EU directives, Berne Convention and WIPO treaties.
- Work vs Idea: Copyright protects the individual expression of ideas (the particular form, text, source code as written), not the underlying ideas, procedures, methods, algorithms or mathematical concepts. Implementations (code) can be protected as expression, the abstract algorithm cannot.
- Computer Programs are treated as literary works for copyright purposes, their source and object code are protected as expression, unprotected elements include algorithms, architectures, and interfaces insofar as they are functional ideas.

Author, Co-author and employer rights:

- Author = creator, Co-authors share copyright and are presumed to hold equal shares unless a contract or agreement states otherwise.
- Employer vs author defaults: Work created by an employee in the course of employment generally gives economic rights to the employer (unless the employment contract says otherwise). Moral (personal) rights (such as attribution and integrity) remain with the author and are inalienable.
- Agreements matter: Transfer of economic rights, scope of license, remuneration and fields of exploration should be clearly written. Unclear or missing clauses leave many rights with the author.

Licences, transfers and fields of exploration:

- A licence or transfer must specify fields of exploitation (media, distribution methods, territory, language, duration). Broad or vague transfers risk being narrowed by courts.
- Unknown future fields cannot be validly granted at the time of contract. You cannot validly transfer an unforeseeable type of exploitation without express agreement.
- If a license omits duration, it may be treated as indefinite but terminable under statutory rules or by contract terms.
- Transfer of a physical copy (sale of a disk) does not equal transfer of the underlying economic copyright (modify, copy it, ...).

Duration of rights and post-term:

- Economic rights term: Generally 70 years after the author's death (subject to special rules for anonymous works, cinematographic works, and co-authorship). After expiry, works enter in public domain but reproduction rules and certain fees (in specific systems) may remain relevant.

Permitted uses and educational exceptions:

- Permitted personal use: "The scope of personal use includes the use of single copies of work by a circle of people who are in a personal relationship, in particular by way of kinship, affinity or social relationship".
Personal, non-commercial copying for a closed circle is allowed.

- Library, educational, and scientific exceptions: Limited reproduction and lending rights apply for libraries, schools and research under specific conditions as non-commercial, limited fragments or single copies, proper attribution, ...
Quotation is allowed only within limits that do not harm normal exploitation.

- Crucial Limit: Permitted/educational exceptions must not conflict with normal exploitation or the legitimate interests of the rightholder.

Criminal and civil liabilities:

- Criminalized conduct includes: unauthorized distribution/reproduction (especially when for profit or organized), circumvention of technological protection measures, producing or distributing devices meant to remove protections, unauthorized access to systems, data destruction or interference with computer systems.
- Business risks: Civil claims for infringement, damages and injunction.
Criminal sanctions where intent, profit motive or organized activity is present.
- Evidence and Enforcement: Seizure, Forensics, and cooperation with enforcement agencies are covered in lectures as typical remedies.

Industrial Property and patentability limits:

- Patentable subject-matter: inventions that are new, involve inventive step, and are industrially applicable.
Pure software, mathematical methods and abstract algorithms are generally excluded from patentability on their own, patentability depends on technical contribution and implementation.
- Other IP types: utility models (short-term protection for technical solutions), industrial designs (appearance), trademarks (distinctive signs), and integrated circuit topographies. Each has separate registration requirements and protection regimes.

Ethics, risks and professional responsibility:

- Codes: IEEE/ACM Software Engineering Code of Ethics and national associations (in Poland: PTI) set duties:
Protect public interest, Deliver safe and reliable products, Be honest with clients and employers, Avoid conflicts of interest, Respect IP and privacy, and maintain professional competence. (8 basic principles)
- Risk management: recognize technical, legal, financial, reputation risks. Use quantitative/qualitative methods to estimate and mitigate them.

Summary Reminders:

- Expression (text, code) = copyrighted / Algorithms and ideas = NOT copyrighted.
- Personal-use → Personal, non-commercial copying for a closed circle is allowed.
- Employer commonly holds economic rights for employee-created software. Moral rights stay with the author.
- Licenses must specify fields of exploitation.
- Unauthorized circumvention/distribution may lead to criminal penalties.
- Economic copyright term ≈ 70 years after author's death.